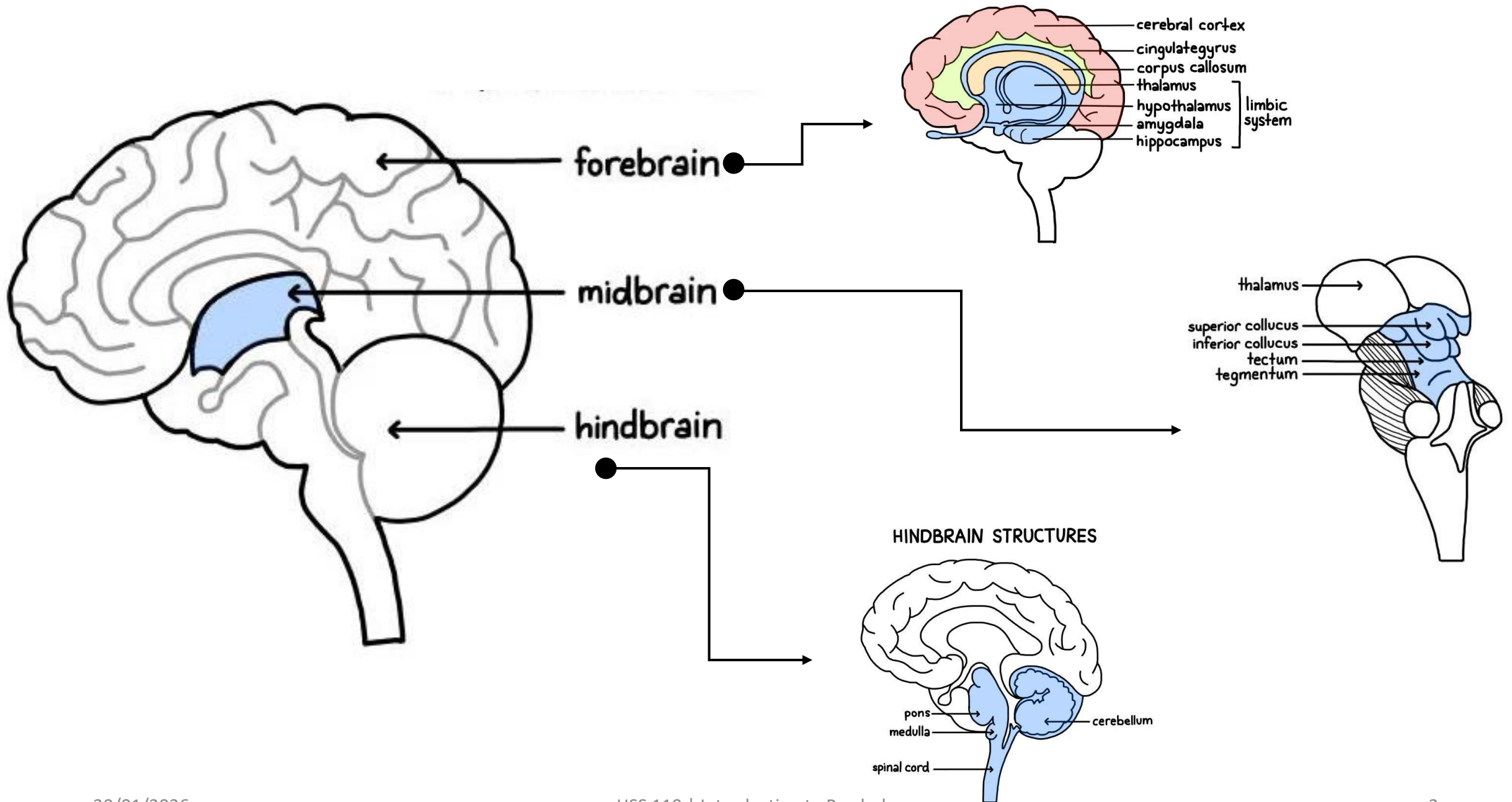




Biological Bases of Behaviour

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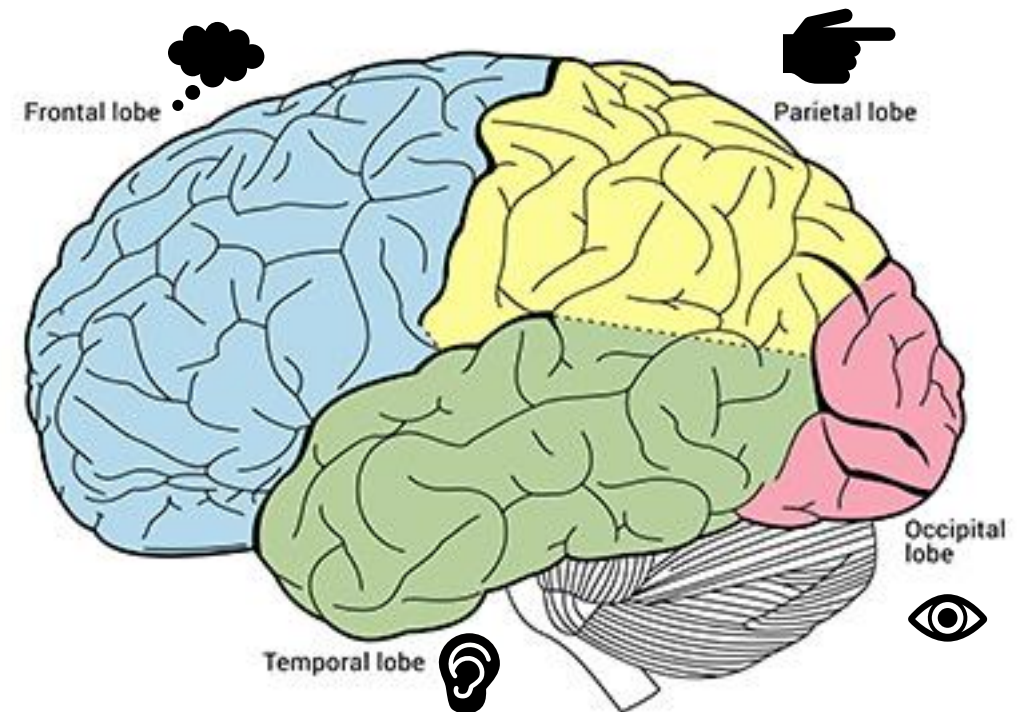
Structure of the Brain: The Forebrain

Cortex, the basal ganglia, and the limbic system

- Cortex: Outermost part of the brain
- Divided into two sections called the cerebral hemispheres: right & left
- Connected by corpus callosum
- Each hemisphere is divided into four lobes or sections

Specialisation of the two hemispheres

Left Hemisphere	Right Hemisphere
Controls the right hand	Controls the left hand
Spoken language	Nonverbal
Written language	Visual-spatial perception
Mathematical calculations	Music and artistic processing
Logical thought processes	Emotional thought and recognition
Analysis of detail	Processes the whole
Reading	Pattern recognition
	Facial recognition



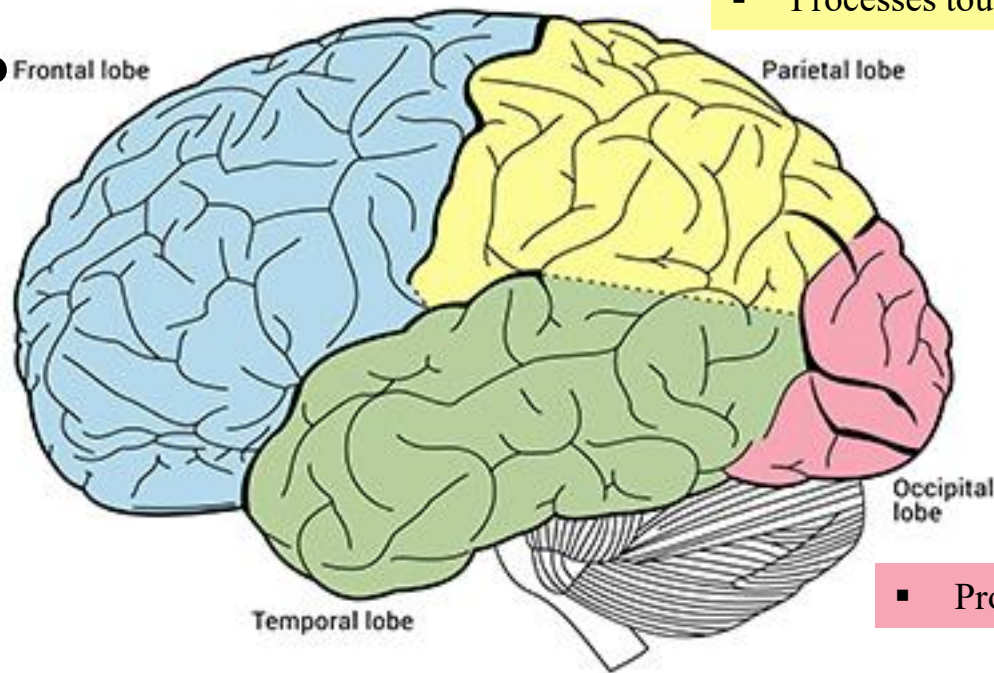
Structure of the Brain: The Forebrain

Cortex, the basal ganglia, and the limbic system

- Processes higher mental functions
- E.g., planning, personality, memory storage, complex decision making, language
- Helps control emotions through the limbic system

Damage: motor tasks

E.g., *perseveration*: repeating the same step, wrong answer, or movement over and over



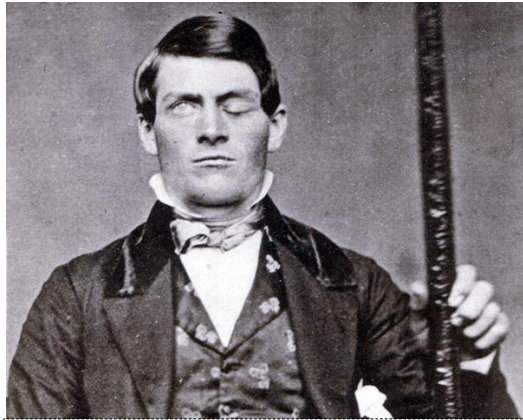
- Processes touch, temperature, and body position

- Processes visual information from the eyes

- Processes hearing and meaningful speech

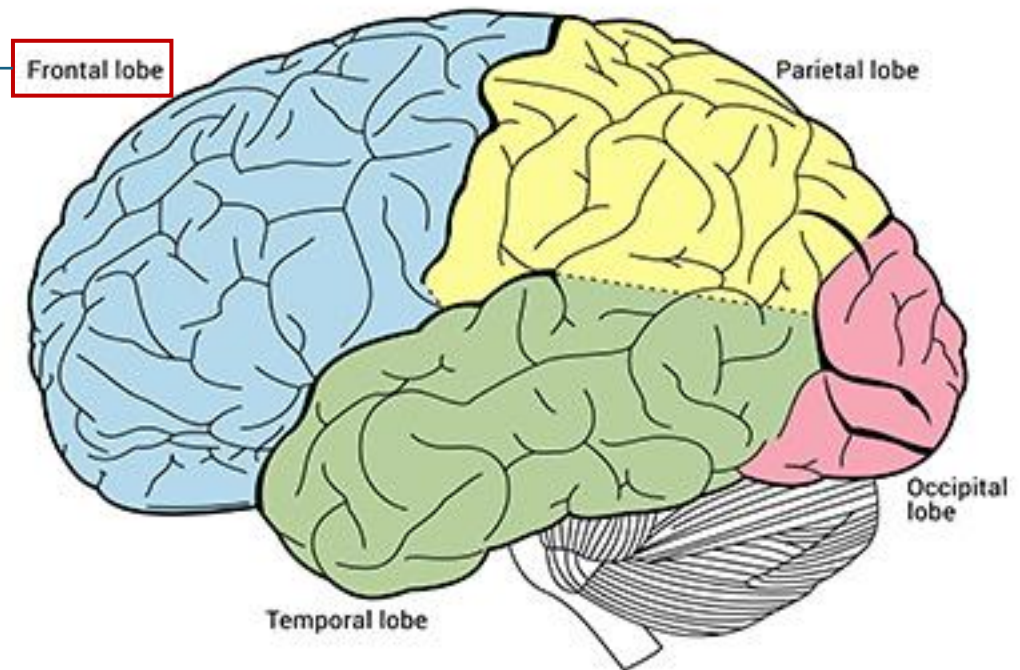
Structure of the Brain: The Forebrain

Cortex, the basal ganglia, and the limbic system



- Changes after injury
- Loss of emotional control
- Change in personality
- Impulsive
- Poor judgment and decision making

Phineas Gage



Structure of the Brain: The Forebrain

Cortex, the basal ganglia, and the limbic system

- Left frontal lobe: Production of speech
- Broca's area: articulation and fluency of speech
- **Broca's aphasia**: expressive language disorder
- Affects fluency but not comprehension
- E.g., "Want water now" instead of "I want a glass of water"

"Dog dog park" instead of "There are two dogs at the park"



Broca = **B**roken speech

Structure of the Brain: The Forebrain

Cortex, the basal ganglia, and the limbic system

- Left temporal lobe
- Wernicke's area: comprehension of words
- **Wernicke's aphasia**: impaired language comprehension
- Fluent speech that does not make sense
- E.g., you mean to say “chair,” but you say “table”

you mean to say “cat,” but you say “hat”



TactusTherapy

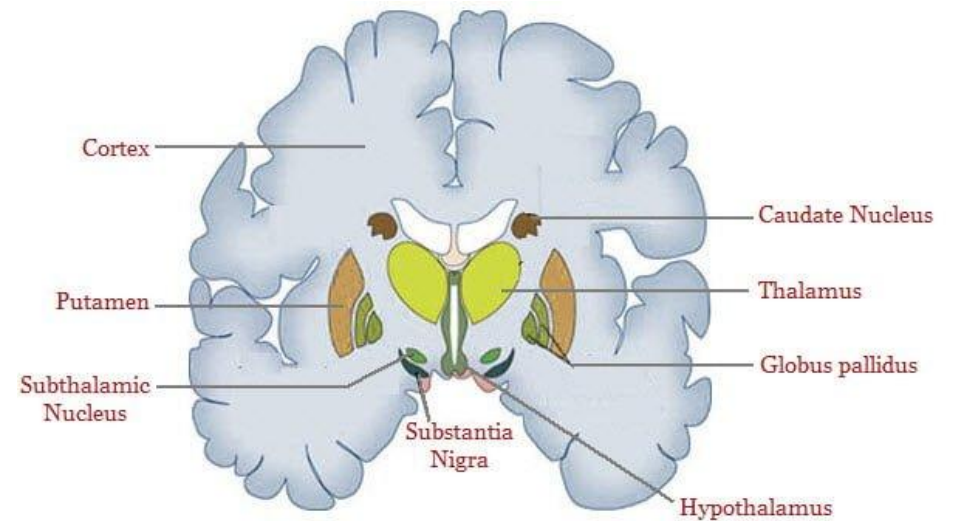


Wernicke = Word salad

Structure of the Brain: The Forebrain

Cortex, the basal ganglia, and the limbic system

- Group of deep brain structures called basal ganglia (or basal nuclei)
- Spread across the brain
- Regulate **voluntary motor movements** by fine-tuning and coordinating actions
- **Involved in decision-making** (evaluates risks and goals), emotions, motivation, learning habits, rewards
- Dysfunction linked to disorders like **addiction, Alzheimer's, Parkinson's, Huntington's** (Decay of brain nerve cells over time), and **Tourette syndrome**



Structure of the Brain: The Forebrain

Cortex, the basal ganglia, and the limbic system

Thalamus

- Relay station
- Hearing, sight, touch, or taste
- Smell is the only sense that does not have to first pass through the thalamus
- Associated with dyslexia

Cingulate cortex

- Active during a variety of cognitive tasks such as selective attention, written word recognition, and working memory
- Associated with attention-deficit/hyperactivity disorder, major depressive disorder, and bipolar disorder

Hypothalamus

- Body temperature, thirst, hunger, sleeping and waking, sexual activity, and emotions
- Controls the pituitary gland, so hormone regulation ultimately depends on it
- Hormones influences: Depression

Hippocampus

- ACh, the neurotransmitter involved in muscle control, is also involved in the memory function
- People who have Alzheimer's have much lower levels of ACh

Amygdala

- Fear responses and memory of fear
- Sensory information reaches the amygdala first, enabling quick responses to danger before conscious awareness
- Damage: decreased fear response

Structure of the Brain: The Forebrain

Cortex, the basal ganglia, and the limbic system

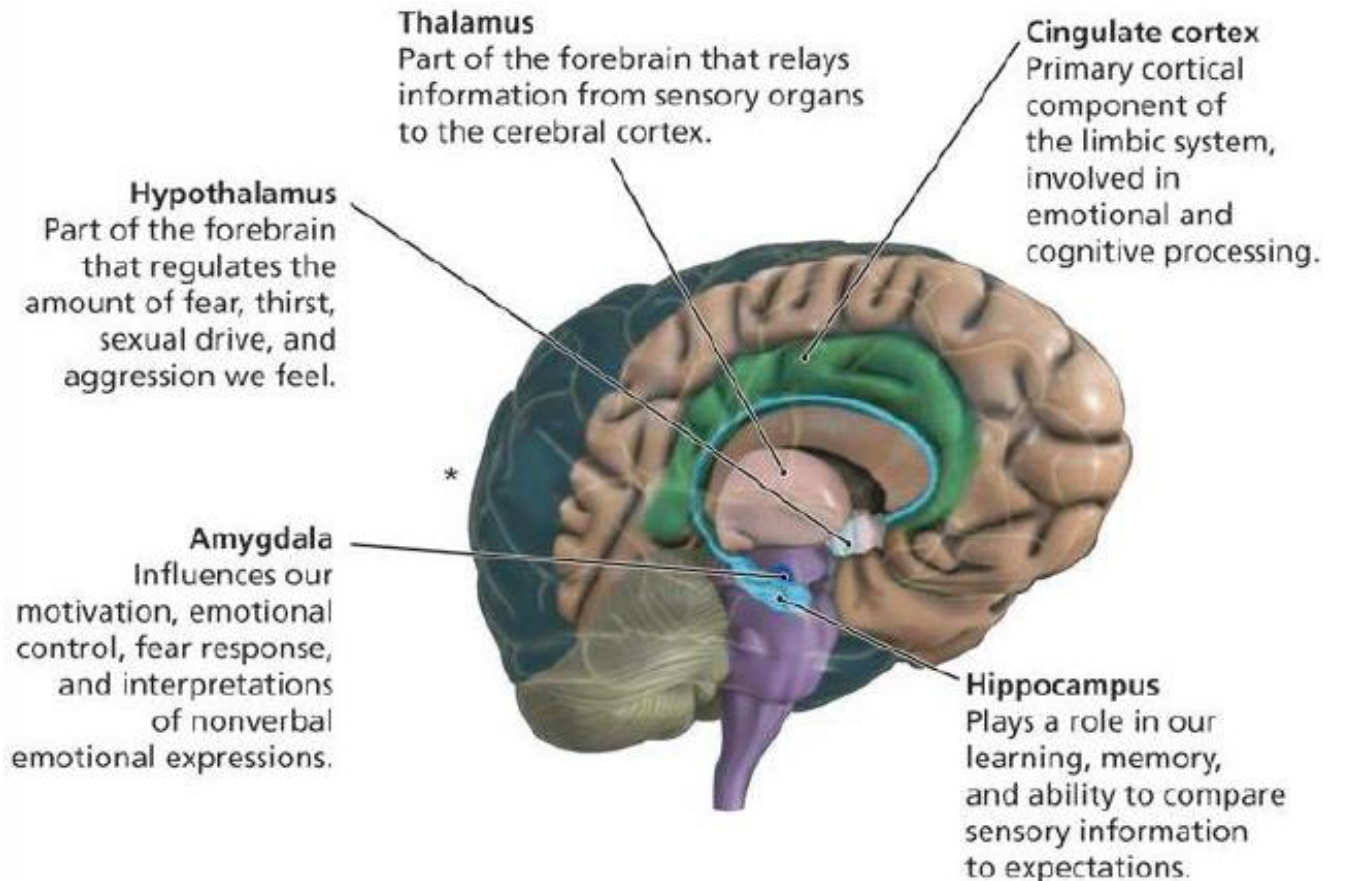
Emotions, motivation, memory, and learning

- **Klüver-Bucy syndrome**

(Neuropsychiatry disorder):

temporal lobe and amygdala

- Memory loss and behavioral issues
- Intense sex drive, compulsive eating, eating inedible objects, dementia etc.



RECAP

- Cortex
 - Frontal lobe: Broca Aphasia (difficulty in using expressive language)
 - Temporal lobe: Wernicke's aphasia (word salad/ difficult in word comprehension)
- Basal Ganglia (decision making and coordination)
 - Addiction
 - Alzheimer's
 - Parkinson's
 - Huntington's (Decay of brain nerve cells over time)
 - Tourette syndrome
- Limbic System
 - Thalamus (relay station for sensory except smell): Dyslexia
 - Hypothalamus (controls emotions): Hormones issues/ Depression
 - Cingulate cortex: Dyslexia (selective attention)
 - Hippocampus: Alzheimer (memory)
 - Amygdala: Klüver-Bucy syndrome